

Lesson 6. Vector calculus I

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Motivation

Consider the differential equation for a simple harmonic oscillator, which can be modeled as:

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$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

This second-order differential equation can be transformed into a system of first-order equations:

$$\frac{dx}{dt} = v, \quad \frac{dv}{dt} = -\omega^2 x$$

Here, x is the position, v is the velocity, and ω is a constant (angular frequency). The corresponding **vector field** is given by:

$$\mathbf{F}(x, v) = \langle v, -\omega^2 x \rangle$$

In this case, the vector field describes how the position and velocity of the oscillator change over time. The trajectories of this vector field correspond to the motion of the oscillator, illustrating its oscillatory behavior.

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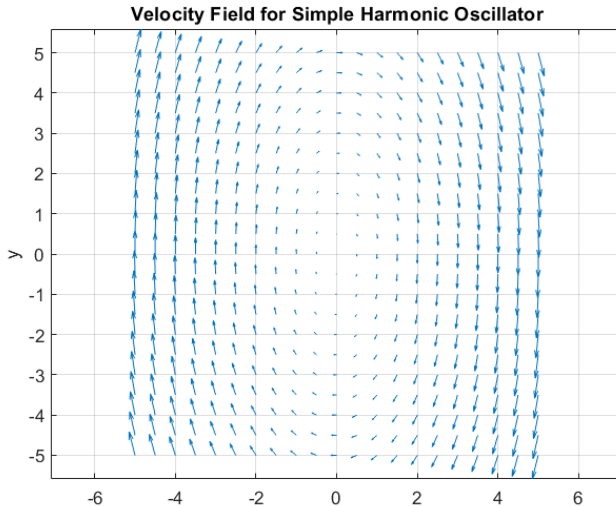
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Velocity field



Definition of VF

A **vector field** on two (or three) dimensional space is a function $\vec{F}$ that assigns to each point (x, y) (or (x, y, z)) a two (or three dimensional) vector given by $\vec{F}(x, y)$ (or $\vec{F}(x, y, z)$).

A vector field can be represented in either of two equivalent ways.

The first way is to use a vector with components that are multi-variable functions:

$$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$$

The second way is to use the standard unit vectors:

$$\vec{F}(x, y) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$$

The functions P , Q , R (if it is present) are sometimes called **scalar functions**.

A vector field is said to be **continuous** if its component functions are continuous.

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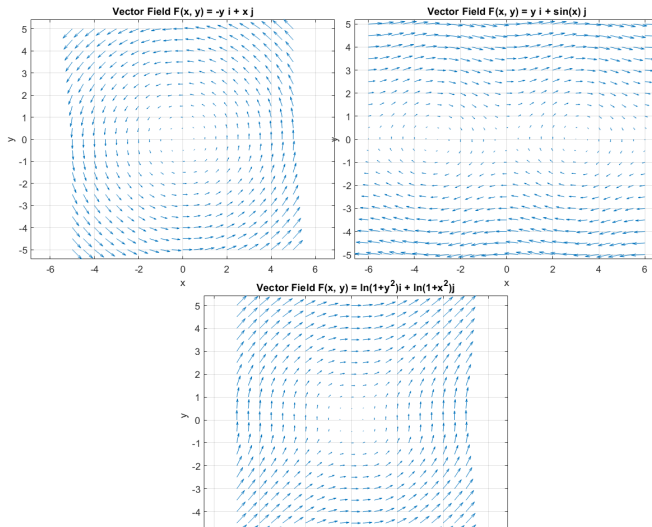
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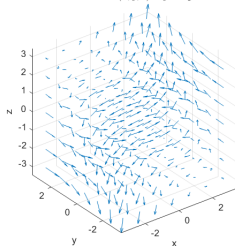
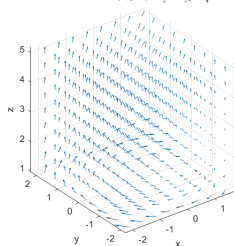
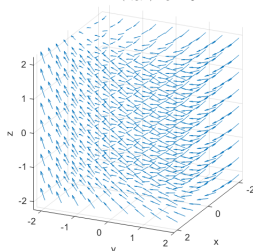
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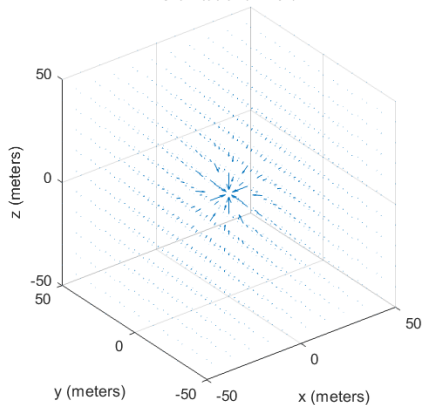
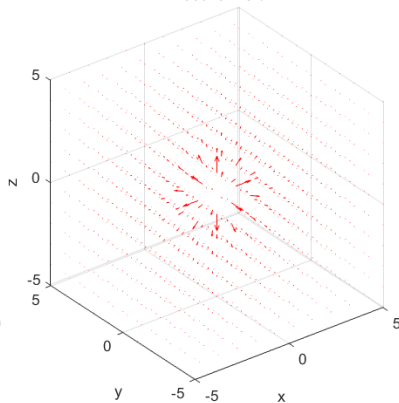
Examples of vector field in 2D



Examples of vector field in 3D

Vector Field $\mathbf{F}(x, y, z) = yi + zj + zk$ Vector Field $\mathbf{F}(x, y, z) = \frac{y}{z}i - \frac{x}{z}j + \frac{z}{4}k$ Vector Field $\mathbf{F}(x, y, z) = yi - 2j + zk$ 

Examples of Force Fields

Gravitational Field**Electric Field**

Grad and Gradient Fields

If f is a (scalar) function of x , y , and z , then the **gradient** of f is

$$\text{grad } f = \nabla f(x, y, z) = f_x(x, y, z)\hat{i} + f_y(x, y, z)\hat{j} + f_z(x, y, z)\hat{k}.$$

This is a vector field and is often called a **gradient vector field**:

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

Find the gradient vector field of the following function:

$$f(x, y) = x^2 \sin(5y)$$

Note that we only gave the gradient vector definition for a three-dimensional function, but don't forget that there is also a two-dimensional definition. All that we need to drop off is the third component of the vector.

Here is the gradient vector field for this function:

$$\nabla f = \langle 2x \sin(5y), 5x^2 \cos(5y) \rangle$$

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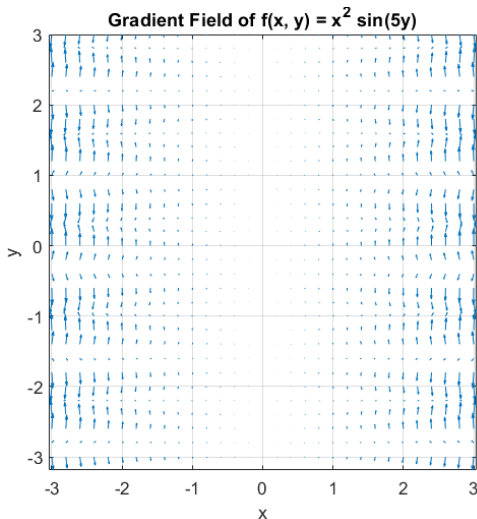
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Example: Gradient vector field



Grad and Gradient Fields

Sketch the gradient vector field for

$$f(x, y) = x^2 + y^2$$

as well as several contours for this function.

The contours are defined by the equation,

$$x^2 + y^2 = k$$

and so they are circles centered at the origin with radius $\sqrt{k}$.

Here is the gradient vector field for this function:

$$\nabla f(x, y) = 2x\hat{i} + 2y\hat{j}$$

Recall that the vectors of the vector field are all orthogonal (or perpendicular) to the contours. This will always be the case when we are dealing with the contours of a function as well as its gradient vector field.

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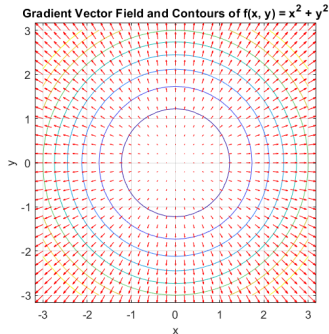
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Example: Gradient vector field and Contours



Notice that the closer the contour curves are (as k is increased by a fixed amount) the faster the function is changing at that point. Also recall that the direction of fastest change for a function is given by the gradient vector at that point. Therefore, it should make sense that the two ideas should match up as they do here.

Conservative vector field

A vector field $\vec{F}$ is called a **conservative vector field** if there exists a function f such that

$$\vec{F} = \nabla f.$$

If $\vec{F}$ is a conservative vector field, then the function f is called a **potential function** for $\vec{F}$.

All this definition is saying is that a vector field is conservative if it is also a gradient vector field for some function:

- the vector field $\vec{F} = y\vec{i} + x\vec{j}$ is a conservative vector field with a potential function of $f(x, y) = xy$ because $\nabla f = \langle y, x \rangle$.
- $\vec{F} = -y\vec{i} + x\vec{j}$ is not a conservative vector field since there is no function f such that $\vec{F} = \nabla f$.

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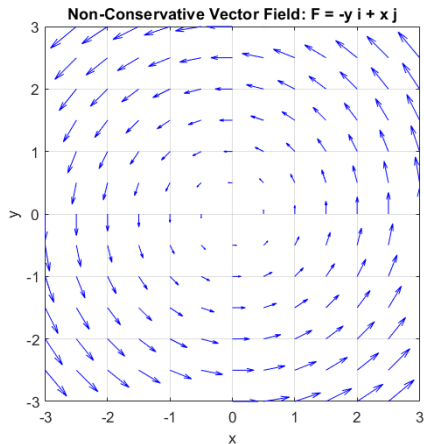
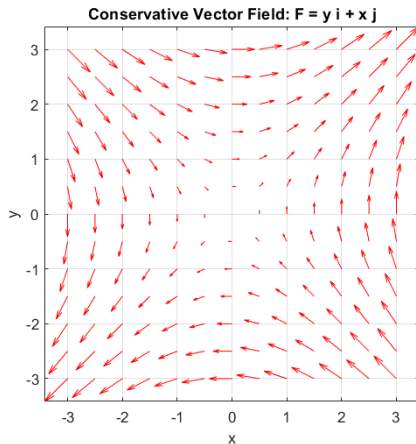
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Theorem (The Cross-Partial Property of Conservative Vector Fields)

Let $\vec{F}$ be a vector field in two or three dimensions such that the component functions of $\vec{F}$ have continuous second-order mixed-partial derivatives on the domain of $\vec{F}$.

If $\vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$ is a conservative vector field in $\mathbb{R}^2$, then

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The Cross-Partial Property of Conservative Vector Fields shows that most vector fields are not conservative. The cross-partial property is difficult to satisfy in general, so most vector fields won't have equal cross-partials.

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Line integrals: introduction

A line integral gives us the ability to integrate multivariable functions and vector fields over arbitrary curves in a plane or in space. There are two types of line integrals: **scalar line integrals** and **vector line integrals**.

Scalar line integrals

Scalar line integrals are integrals of a *scalar function* over a *curve* in a plane or in space.

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Scalar line integrals

Derivation

Let C be a smooth curve in space parameterized by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$ for $a \leq t \leq b$.

To define the line integral of a function $f(x, y, z)$ over C , we partition the interval $[a, b]$ into n subintervals $[t_{i-1}, t_i]$.

Let t_i^* be a point in the i -th interval, and denote the endpoints of the curve as $P_0, P_1, \dots, P_n$. Each segment C_i has length Δs_i .

We evaluate f at P_i^* and multiply by the length Δs_i to form a sum:

$$\sum_{i=1}^n f(P_i^*) \Delta s_i.$$

Taking the limit as $n \rightarrow \infty$ gives the line integral:

$$\int_C f \, ds.$$

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Scalar line integrals: Video Demonstration

https://en.wikipedia.org/wiki/Line_integral

Scalar line integrals

Theorem (Scalar Line Integral Calculation)

Let f be a continuous function with a domain that includes the smooth curve C with parameterization $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$. Then

$$\begin{aligned}\int_C f(x, y, z) \, ds &= \int_a^b f(x(t), y(t), z(t)) \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \, dt = \\ &= \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| \, dt\end{aligned}$$

Similarly,

$$\int_C f(x, y) \, ds = \int_a^b f(x(t), y(t)) \sqrt{(x'(t))^2 + (y'(t))^2} \, dt = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| \, dt$$

if C is a planar curve and f is a function of two variables.

Parameterization of the Basic Curves

Curve	Parametric Equations	
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (Ellipse)	Counter-Clockwise	Clockwise
	$x = a \cos(t)$ $y = b \sin(t)$ $0 \leq t \leq 2\pi$	$x = a \cos(t)$ $y = -b \sin(t)$ $0 \leq t \leq 2\pi$
$x^2 + y^2 = r^2$ (Circle)	Counter-Clockwise	Clockwise
	$x = r \cos(t)$ $y = r \sin(t)$ $0 \leq t \leq 2\pi$	$x = r \cos(t)$ $y = -r \sin(t)$ $0 \leq t \leq 2\pi$
$y = f(x)$	$x = t$ $y = f(t)$	
$x = g(y)$	$x = g(t)$ $y = t$	
Line Segment	$\vec{r}(t) = (1-t)\langle x_0, y_0, z_0 \rangle + t\langle x_1, y_1, z_1 \rangle, \quad 0 \leq t \leq 1$	
From (x_0, y_0, z_0) to (x_1, y_1, z_1)	or $x = (1-t)x_0 + tx_1$ $y = (1-t)y_0 + ty_1$ $z = (1-t)z_0 + tz_1$ $\quad \quad \quad , \quad 0 \leq t \leq 1$	

Scalar line integrals

Example 1: Evaluate

$$\int_C xy^4 ds$$

where C is the right half of the circle, $x^2 + y^2 = 16$, traced out in a counterclockwise direction.

Solution

We first need a parameterization of the circle. This is given by,

$$x = 4\cos t \quad \text{and} \quad y = 4\sin t$$

We now need a range of t 's that will give the right half of the circle. The following range of t 's will do this:

$$-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

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Scalar line integrals

Solution

Now, we need the derivatives of the parametric equations and compute ds .

$$x'(t) = \frac{dx}{dt} = -4 \sin t \quad \text{and} \quad y'(t) = \frac{dy}{dt} = 4 \cos t$$

$$ds = \sqrt{16 \sin^2 t + 16 \cos^2 t} dt = 4 dt$$

The line integral is then,

$$\int_C xy^4 ds = \int_{-\pi/2}^{\pi/2} 4 \cos t \cdot (4 \sin t)^4 \cdot 4 dt = 4096 \int_{-\pi/2}^{\pi/2} \cos t \sin^4 t dt =$$

This evaluates to:

$$= 4096 \cdot \frac{1}{5} \sin^5 t \Big|_{-\pi/2}^{\pi/2} = \frac{8192}{5}$$

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Scalar line integrals: Properties

The line integrals of piecewise smooth curves are the sum of the line integrals evaluated over each of the pieces:

$$\int_C f(x, y) ds = \int_{C_1} f(x, y) ds + \int_{C_2} f(x, y) ds + \int_{C_3} f(x, y) ds + \dots + \int_{C_n} f(x, y) ds$$

The line integrals with respect to arc length do not depend on the direction of the curve and do not change the value of the integral:

$$\int_C f(x, y) ds = \int_{-C} f(x, y) ds$$

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Scalar line integrals

Example 2: Evaluate

$$\int_C 4x^3 ds$$

where C is the line segment from $(-2, -1)$ to $(1, 2)$.

Solution:

From the parameterization formulas in the table, we know that the line segment starting at $(-2, -1)$ and ending at $(1, 2)$ is given by:

$$\mathbf{r}(t) = (1-t)\langle -2, -1 \rangle + t\langle 1, 2 \rangle = \langle -2+3t, -1+3t \rangle \quad \text{for } 0 \leq t \leq 1.$$

This means that the individual parametric equations are:

$$x = -2 + 3t, \quad y = -1 + 3t.$$

Using this path, the line integral is:

$$\int_C 4x^3 ds = \int_0^1 4(-2+3t)^3 \sqrt{9+9} dt = \int_0^1 4(-2+3t)^3 \cdot 3 dt.$$

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Scalar line integrals

Now, simplifying further:

$$= 12\sqrt{2} \int_0^1 (-2 + 3t)^3 dt = 12\sqrt{2} \left(\frac{1}{12} (-2 + 3t)^4 \Big|_0^1 \right).$$

Evaluating the integral:

$$\begin{aligned} &= 12\sqrt{2} \left(\frac{1}{12} \right) ((-2 + 3(1))^4 - (-2 + 3(0))^4) = 12\sqrt{2} \left(\frac{1}{12} \right) (1^4 - (-2)^4) . \\ &= 12\sqrt{2} \left(\frac{1}{12} \right) (1 - 16) = 12\sqrt{2} \left(\frac{1}{12} \right) (-15) = -15\sqrt{2} \approx -21.213. \end{aligned}$$

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where C is the line segment from $(1, 2)$ to $(-2, -1)$.

Solution:

Here is the parameterization of the curve:

$$\mathbf{r}(t) = (1 - t)\langle 1, 2 \rangle + t\langle -2, -1 \rangle = \langle 1 - 3t, 2 - 3t \rangle \quad \text{for } 0 \leq t \leq 1.$$

Remember that we are switching the direction of the curve, and this will also change the parameterization, so we can make sure that we start/end at the proper point.

Here is the line integral:

$$\int_C 4x^3 ds = \int_0^1 4(1 - 3t)^3 \sqrt{9 + 9} dt = 12\sqrt{2} \int_0^1 (1 - 3t)^3 dt.$$

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Continuing with the calculations:

$$= 12\sqrt{2} \left(\frac{1}{12} \right) (16 - 1) = 12\sqrt{2} \left(\frac{1}{12} \right) (15) = 15\sqrt{2}.$$

Thus, the final result is:

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Scalar line integrals

Up to now, we have been looking at line integrals with respect to arc length. Now, we would like to look at line integrals with respect to x and/or y .

The **line integral of f with respect to x** is,

$$\int_C f(x, y) dx = \int_a^b f(x(t), y(t)) x'(t) dt$$

The **line integral of f with respect to y** is,

$$\int_C f(x, y) dy = \int_a^b f(x(t), y(t)) y'(t) dt$$

These two integrals often appear together, and so we have the following shorthand notation for these cases:

$$\int_C P dx + Q dy = \int_C P(x, y) dx + \int_C Q(x, y) dy$$

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$$\int_C f(x, y, z) \, dz = \int_a^b f(x(t), y(t), z(t)) z'(t) \, dt$$

where the curve C is parameterized by

$$x = x(t) \quad y = y(t) \quad z = z(t) \quad a \leq t \leq b$$

As with the two-dimensional version these three will often occur together so the shorthand we'll be using here is

$$\int_C P dx + Q dy + R dz = \int_C P(x, y, z) \, dx + \int_C Q(x, y, z) \, dy + \int_C R(x, y, z) \, dz$$

Scalar line integrals

If C is any curve then,

$$\int_{-C} P dx + Q dy = - \int_C P dx + Q dy$$

or

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Example 4: Evaluate:

$$\int_C x^3 dy - (y + 1) dx$$

for each of the following curves:

- ① C is the line segment from $(1, 7)$ to $(-2, 4)$.
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Scalar line integrals

Solution (1)

For reasons that will become apparent once we get to the second part of this problem, let's call this curve C_1 instead of C . Here then is the parameterization of C_1 .

$$C_1 : \vec{r}(t) = (1-t)\langle 1, 7 \rangle + t\langle -2, 4 \rangle = \langle 1-3t, 7-3t \rangle, \quad 0 \leq t \leq 1$$

Here is the line integral for this curve.

$$\int_{C_1} x^3 dy - (y+1) dx = \int_{C_1} x^3 dy - \int_{C_1} (y+1) dx$$

Substitute the parametric equations and differentials:

$$\begin{aligned} &= \int_0^1 (1-3t)^3(-3) dt - \int_0^1 (7-3t+1)(-3) dt \\ &= \int_0^1 -3(1-3t)^3 + 3(8-3t) dt \end{aligned}$$

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Solution (1)

Now, integrate:

$$= \left[\frac{1}{4}(1-3t)^4 + 24t - \frac{9}{2}t^2 \right]_0^1$$

Evaluating at the limits $t = 0$ and $t = 1$, we get:

$$= \frac{93}{4}$$

Solution (2)

Now, as we did in the previous part let's "rename" this curve as C_2 instead of C . Next, note that this curve is just the curve from the first step with opposite direction. In other words what we have here is that $C_2 = -C_1$.

$$\int_{C_2} x^3 dy - (y+1) dx = \int_{-C_1} x^3 dy - (y+1) dx = - \int_{C_1} x^3 dy - (y+1) dx = -\frac{93}{4}$$

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Vector line integrals

The second type of line integrals are **vector line integrals**, in which we integrate along a curve through a vector field.

For example, let

$$\vec{F}(x, y, z) = P(x, y, z)\hat{i} + Q(x, y, z)\hat{j} + R(x, y, z)\hat{k}$$

be a continuous vector field in $\mathbb{R}^3$ that represents a force on a particle, and let C be a smooth curve in $\mathbb{R}^3$ contained in the domain of $\vec{F}$.

How would we compute the work done by $\vec{F}$ in moving a particle along C ?

To answer this question, first note that a particle could travel in two directions along a curve:

- a forward direction;
- a backward direction.

The work done by the vector field depends on the direction in which the particle is moving.

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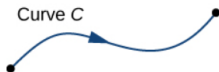
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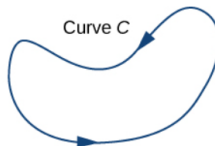
Vector line integrals

Therefore, we must specify a direction along curve C

- Such a specified direction is called an **orientation** of a curve.
- The specified direction is the **positive direction** along C ; the opposite direction is the **negative direction** along C .
- When C has been given an orientation, C is called an **oriented curve**.
- A **closed curve** is one for which there exists a parameterization $\vec{r}(t)$, $a \leq t \leq b$, such that $\vec{r}(a) = \vec{r}(b)$, and the curve is traversed exactly once. In other words, the parameterization is one-to-one on the domain (a, b) .



(a) Oriented curve

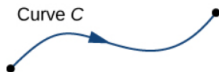


(b) Closed curve

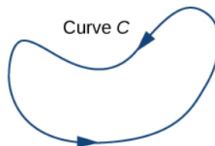
Vector line integrals

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- Such a specified direction is called an **orientation** of a curve.
- The specified direction is the **positive direction** along C ; the opposite direction is the **negative direction** along C .
- When C has been given an orientation, C is called an **oriented curve**.
- A **closed curve** is one for which there exists a parameterization $\vec{r}(t)$, $a \leq t \leq b$, such that $\vec{r}(a) = \vec{r}(b)$, and the curve is traversed exactly once. In other words, the parameterization is one-to-one on the domain (a, b) .



(a) Oriented curve

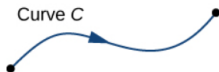


(b) Closed curve

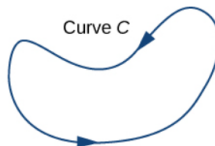
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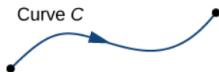


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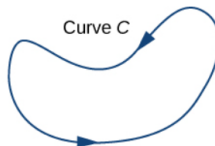
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(b) Closed curve

Vector line integrals

Derivation

Let $\vec{r}(t)$ be a parameterization of the curve C for $a \leq t \leq b$, ensuring that the particle traverses the curve exactly once in the positive direction.

Divide the parameter interval $[a, b]$ into n subintervals $[t_{i-1}, t_i]$, where the points $P_0, P_1, \dots, P_n$ represent the endpoints of the curve segments, effectively dividing C into n pieces.

For each segment, the length is denoted by Δs_i , and a value t_i^* is chosen in each subinterval.

The particle approximately moves in the direction of the unit tangent vector $\vec{T}(P_i)$ at the point P_i^* , which is the endpoint of $\vec{r}(t_i^*)$.

The total work done by the force field along the curve is computed by summing the contributions

$$\sum_{i=1}^n \vec{F}(P_i^*) \cdot \vec{T}(P_i^*) \Delta s_i.$$

Vector line integrals: Video Demonstration

https://en.wikipedia.org/wiki/Line_integral

Vector line integrals

The **vector line integral** of vector field $\vec{F}$ along oriented smooth curve C is

$$\int_C \vec{F} \cdot \vec{T} ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}(P_i^*) \cdot \vec{T}(P_i^*) \Delta s_i$$

if that limit exists.

With scalar line integrals, neither the orientation nor the parameterization of the curve matters. With vector line integrals, the orientation of the curve does matter: reversing the path changes the work value from negative to positive and vice versa. As with scalar line integrals, it is easier to compute a vector line integral if we express it in terms of the parameterization function $\vec{r}$ and the variable t .

Theorem (Evaluating a vector line integral). Let $\vec{F}$ be continuous vector field with domain that include the oriented smooth curve C with parameterization $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$. Then

$$\int_C \vec{F} \cdot \vec{T} ds = \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \quad (1)$$

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Vector line integrals

Given the vector field $\vec{F}(x, y, z) = P\hat{i} + Q\hat{j} + R\hat{k}$ and the curve C parameterized by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $a \leq t \leq b$, the line integral is

$$\int_C \vec{F} \cdot d\vec{r} = \int_C P dx + Q dy + R dz. \quad (2)$$

Theorem: Properties of Vector Line Integrals

Let $\vec{F}$ and $\vec{G}$ be continuous vector fields with domains that include the oriented smooth curve C . Then:

- 1 $\int_C (\vec{F} + \vec{G}) \cdot d\vec{r} = \int_C \vec{F} \cdot d\vec{r} + \int_C \vec{G} \cdot d\vec{r};$
- 2 $\int_C (k\vec{F}) \cdot d\vec{r} = k \int_C \vec{F} \cdot d\vec{r};$
- 3 $\int_{-C} \vec{F} \cdot d\vec{r} = - \int_C \vec{F} \cdot d\vec{r};$
- 4 Suppose instead that C is a piecewise smooth curve in the domains of $\vec{F}$ and $\vec{G}$, where $C = C_1 + C_2 + \cdots + C_n$ and $C_1, C_2, \dots, C_n$ are smooth curves such that the endpoint of C_i is the starting point of C_{i+1} . Then:

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Vector line integrals

Example: Evaluating a Vector Line Integral

Find the value of the integral

$$\int_C \vec{F} \cdot d\vec{r}$$

where C is the semicircle parameterized by $\vec{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq \pi$, and $\vec{F} = \langle -y, x \rangle$.

Solution:

We can use Equation (1) to convert the variable of integration from s to t . We then have:

$$\vec{F}(\vec{r}(t)) = \langle -\sin t, \cos t \rangle \quad \text{and} \quad \vec{r}'(t) = \langle -\sin t, \cos t \rangle.$$

Therefore,

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^\pi \langle -\sin t, \cos t \rangle \cdot \langle -\sin t, \cos t \rangle dt = \int_0^\pi \sin^2 t + \cos^2 t dt = \int_0^\pi 1 dt = \pi.$$

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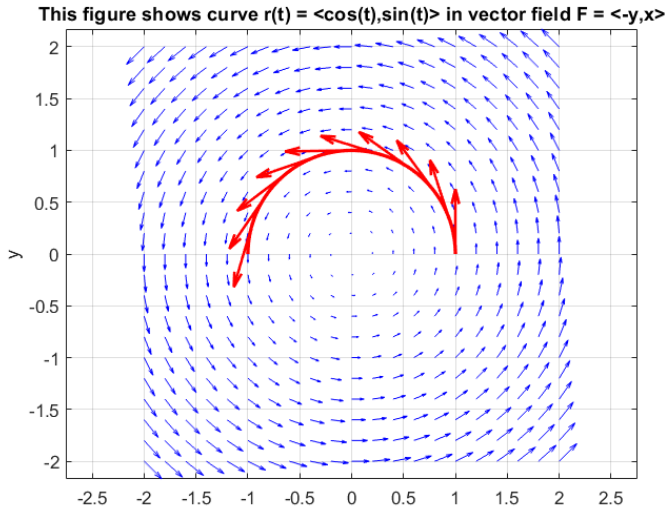
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Vector line integrals



Vector line integrals

Example: Finding the Value of an Integral of the Form $\int_C P dx + Q dy + R dz$

Find the value of the integral

$$\int_C z dx + x dy + y dz,$$

where C is the curve parameterized by $\mathbf{r}(t) = \langle t^2, \sqrt{t}, t \rangle$, $1 \leq t \leq 4$.

Solution:

As with our previous examples, to compute this line integral, we perform a change of variables to express everything in terms of t . Using the parameterization and applying Equation (2), we proceed as follows:

$$\begin{aligned} \int_C z dx + x dy + y dz &= \int_1^4 \left(t(2t) + t^2 \left(\frac{1}{2\sqrt{t}} \right) + \sqrt{t} \right) dt \\ &= \int_1^4 \left(2t^2 + \frac{t^{3/2}}{2} + \sqrt{t} \right) dt. \end{aligned}$$

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Vector line integrals

Now, integrating each term:

$$= \left[\frac{2t^3}{3} + \frac{2t^{5/2}}{5} + \frac{2t^{3/2}}{3} \right]_{t=1}^{t=4}.$$

Substituting the limits $t = 4$ and $t = 1$:

$$\begin{aligned} &= \left(\frac{2(4)^3}{3} + \frac{2(4)^{5/2}}{5} + \frac{2(4)^{3/2}}{3} \right) - \left(\frac{2(1)^3}{3} + \frac{2(1)^{5/2}}{5} + \frac{2(1)^{3/2}}{3} \right) \\ &= \left(\frac{128}{3} + \frac{64}{5} + \frac{16}{3} \right) - \left(\frac{2}{3} + \frac{2}{5} + \frac{2}{3} \right) \\ &= \frac{1440}{15} + \frac{192}{15} + \frac{80}{15} - \left(\frac{10}{15} + \frac{6}{15} + \frac{10}{15} \right) \\ &= \frac{1712}{15} - \frac{26}{15} = \frac{1686}{15} = 112.4. \end{aligned}$$

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Example: Using Properties to Compute a Vector Line Integral

Find the value of the integral

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds,$$

where C is the rectangle (oriented counterclockwise) in the plane with vertices $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(0, 1)$, and where $\mathbf{F} = \langle x - 2y, y - x \rangle$.

Solution

We compute this line integral by breaking the curve C into four segments corresponding to each side of the rectangle, and summing the line integrals over each segment.

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Vector line integrals

Segment 1: From $(0, 0)$ to $(2, 0)$

On this segment, $y = 0$, so the parameterization is $\mathbf{r}_1(t) = \langle t, 0 \rangle$ for $0 \leq t \leq 2$. The tangent vector is $\mathbf{r}'_1(t) = \langle 1, 0 \rangle$, and the vector field $\mathbf{F}$ becomes:

$$\mathbf{F}(t, 0) = \langle t, -t \rangle.$$

Thus, the line integral on this segment is:

$$\int_0^2 \mathbf{F}(t, 0) \cdot \mathbf{r}'_1(t) dt = \int_0^2 t dt = \left[\frac{t^2}{2} \right]_0^2 = 2.$$

Vector line integrals

Segment 2: From $(2, 0)$ to $(2, 1)$

On this segment, $x = 2$, so the parameterization is $\mathbf{r}_2(t) = \langle 2, t \rangle$ for $0 \leq t \leq 1$. The tangent vector is $\mathbf{r}'_2(t) = \langle 0, 1 \rangle$, and the vector field $\mathbf{F}$ becomes:

$$\mathbf{F}(2, t) = \langle 2 - 2t, t - 2 \rangle.$$

Thus, the line integral on this segment is:

$$\int_0^1 \mathbf{F}(2, t) \cdot \mathbf{r}'_2(t) dt = \int_0^1 (t - 2) dt = \left[\frac{t^2}{2} - 2t \right]_0^1 = -\frac{3}{2}.$$

Vector line integrals

Segment 3: From $(2, 1)$ to $(0, 1)$

On this segment, $y = 1$, so the parameterization is $\mathbf{r}_3(t) = \langle 2 - t, 1 \rangle$ for $0 \leq t \leq 2$.

The tangent vector is $\mathbf{r}'_3(t) = \langle -1, 0 \rangle$, and the vector field $\mathbf{F}$ becomes:

$$\mathbf{F}(2 - t, 1) = \langle 2 - t - 2, 1 - (2 - t) \rangle = \langle -t, t - 1 \rangle.$$

Thus, the line integral on this segment is:

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Vector line integrals

Segment 4: From $(0, 1)$ to $(0, 0)$

On this segment, $x = 0$, so the parameterization is $\mathbf{r}_4(t) = \langle 0, 1 - t \rangle$ for $0 \leq t \leq 1$.

The tangent vector is $\mathbf{r}'_4(t) = \langle 0, -1 \rangle$, and the vector field $\mathbf{F}$ becomes:

$$\mathbf{F}(0, 1 - t) = \langle -2(1 - t), 1 - t \rangle.$$

Thus, the line integral on this segment is:

$$\int_0^1 \mathbf{F}(0, 1 - t) \cdot \mathbf{r}'_4(t) dt = \int_0^1 (t - 1) dt = \left[\frac{t^2}{2} - t \right]_0^1 = -\frac{1}{2}.$$

Finally, summing all the contributions, we get the total value of the line integral:

$$\int_C \mathbf{F} \cdot \mathbf{T} ds = 2 + \left(-\frac{3}{2} \right) + 2 - \frac{1}{2} = 2.$$

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Applications of Line Integrals

Scalar line integrals:

They can be used to calculate

- the length or mass of a wire;
- the surface area of a sheet at a given height;
- the electric potential of a charged wire given a linear charge density;

and more.

Vector line integrals:

They can be used to calculate

- the work done on a particle as it moves through a force field;
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Applications of Line Integrals: Calculating the Mass of a Wire

Example: Mass of a Wire

Calculate the mass of a spring in the shape of a curve parameterized by:

$$\mathbf{r}(t) = \langle t, 2 \cos t, 2 \sin t \rangle, \quad 0 \leq t \leq \frac{\pi}{2}$$

with a density function given by:

$$\rho(x, y, z) = e^x + yz \quad \text{kg/m}$$

Applications of Line Integrals: Calculating the Mass of a Wire

Solution:

To calculate the mass of the spring, we must compute the scalar line integral:

$$\int_C \rho(x, y, z) ds$$

where C is the given helix. Using the parameterization $\mathbf{r}(t)$, we write the integral in terms of t .

The differential arc length element ds is given by:

$$ds = \|\mathbf{r}'(t)\| dt$$

Applications of Line Integrals: Calculating the Mass of a Wire

First, compute $\mathbf{r}'(t)$:

$$\mathbf{r}'(t) = \langle 1, -2 \sin t, 2 \cos t \rangle$$

The magnitude of $\mathbf{r}'(t)$ is:

$$\|\mathbf{r}'(t)\| = \sqrt{1^2 + (-2 \sin t)^2 + (2 \cos t)^2} = \sqrt{1 + 4 \sin^2 t + 4 \cos^2 t} = \sqrt{5}$$

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Applications of Line Integrals: Calculating the Mass of a Wire

Now, the density function in terms of t is:

$$\rho(t) = e^t + 4 \cos t \sin t$$

Thus, the mass integral becomes:

$$\int_C \rho(x, y, z) ds = \int_0^{\frac{\pi}{2}} (e^t + 4 \cos t \sin t) \sqrt{5} dt$$

This simplifies to:

$$\sqrt{5} \int_0^{\frac{\pi}{2}} (e^t + 4 \cos t \sin t) dt$$

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Applications of Line Integrals: Calculating the Mass of a Wire

The integral can be split into two parts:

$$\sqrt{5} \left[\int_0^{\frac{\pi}{2}} e^t dt + \int_0^{\frac{\pi}{2}} 4 \cos t \sin t dt \right]$$

The first integral is straightforward:

$$\int_0^{\frac{\pi}{2}} e^t dt = e^{\frac{\pi}{2}} - e^0 = e^{\frac{\pi}{2}} - 1$$

For the second integral, we use the identity $2 \cos t \sin t = \sin(2t)$:

$$4 \int_0^{\frac{\pi}{2}} \cos t \sin t dt = 2 \int_0^{\frac{\pi}{2}} \sin(2t) dt = -\cos(2t) \Big|_0^{\frac{\pi}{2}} = 2$$

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Applications of Line Integrals: Calculating the Mass of a Wire

Thus, the total mass is:

$$\sqrt{5} \left(e^{\frac{\pi}{2}} + 1 \right)$$

Therefore, the mass of the spring is:

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Applications of Line Integrals: Calculating Work

Example: Work Done

How much work is required to move an object in the vector force field $\vec{F} = \langle yz, xy, xz \rangle$ along the path $\vec{r}(t) = \langle t^2, t, t^4 \rangle$, $0 \leq t \leq 1$?

Solution:

Let C denote the given path. We need to find the value of $\int_C \vec{F} \cdot d\vec{r}$. To do this, we use Equation (1):

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 (\langle t^5, t^3, t^6 \rangle \cdot \langle 2t, 1, 4t^3 \rangle) dt = \int_0^1 (2t^6 + t^3 + 4t^9) dt = \\ &= \left[\frac{2t^7}{7} + \frac{t^4}{4} + \frac{2t^{10}}{10} \right]_{t=1}^{t=0} = \frac{13}{1140} \text{ units of work.} \end{aligned}$$

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Applications of Line Integrals: Flux across a plane curve

The **flux** is the rate of a fluid flowing across a curve in a vector field.

The flux of $\vec{F}$ across C is the line integral

$$\int_C \vec{F} \cdot \vec{n}(t) \|\vec{n}(t)\| ds.$$

Calculating Flux Across a Curve: Let $\vec{F}$ be a vector field and let C be a smooth curve with parameterization $r(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$. Let $\vec{n}(t) = \langle y'(t), -x'(t) \rangle$. The flux of $\vec{F}$ across C is

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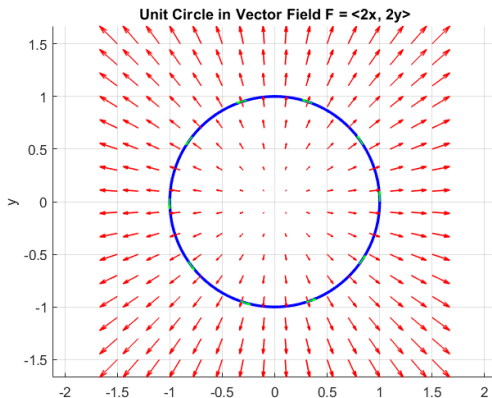
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Applications of Line Integrals: Flux across a plane curve

Example: Flux across a Curve

Calculate the flux of $\vec{F} = \langle 2x, 2y \rangle$ across a unit circle oriented counterclockwise.



Applications of Line Integrals: Flux across a plane curve

Solution:

To compute the flux, we first need a parameterization of the unit circle. We can use the standard parameterization $\vec{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$. The normal vector to a unit circle is $\langle \cos t, \sin t \rangle$. Therefore, the flux is

$$\begin{aligned} \int_C \vec{F} \cdot \vec{N} \, ds &= \int_0^{2\pi} \langle 2 \cos t, 2 \sin t \rangle \cdot \langle \cos t, \sin t \rangle \, dt = \\ &= \int_0^{2\pi} (2 \cos^2 t + 2 \sin^2 t) \, dt = 2 \int_0^{2\pi} (\cos^2 t + \sin^2 t) \, dt = 2 \int_0^{2\pi} dt = 4\pi. \end{aligned}$$

Applications of Line Integrals: Circulation

The **circulation** is the tendency of a fluid to move in the direction of curve C .

If C is a closed curve, then the circulation of $\vec{F}$ along C is the line integral

$$\oint_C \vec{F} \cdot \vec{T} ds$$

The circle on the integral symbol denotes that C is "circular" in that it has no endpoints.

To see where the term circulation comes from and what it measures, let $\vec{v}$ represent the velocity field of a fluid and let C be an oriented closed curve. At a particular point P , the closer the direction of $\vec{v}(P)$ is to the direction of $\vec{T}(P)$, the larger the value of the dot product $\vec{v}(P) \cdot \vec{T}(P)$. The maximum value of $\vec{v}(P) \cdot \vec{T}(P)$ occurs when the two vectors are pointing in the exact same direction; the minimum value of $\vec{v}(P) \cdot \vec{T}(P)$ occurs when the two vectors are pointing in opposite directions. Thus, the value of the circulation $\oint_C \vec{v} \cdot \vec{T} ds$ measures the tendency of the fluid to move in the direction of C .

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Applications of Line Integrals: Circulation

Example: Calculating Circulation

Let $\vec{F} = \langle -y, x \rangle$ be the vector field and let C represent the unit circle oriented counterclockwise. Calculate the circulation of $\vec{F}$ along C .

Solution:

We use the standard parameterization of the unit circle: $\vec{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$. Then, $\vec{F}(\vec{r}(t)) = \langle -\sin t, \cos t \rangle$ and $\vec{r}'(t) = \langle -\sin t, \cos t \rangle$. Therefore, the circulation of $\vec{F}$ along C is

$$\begin{aligned} \oint_C \vec{F} \cdot \vec{T} \, ds &= \int_0^{2\pi} \langle -\sin t, \cos t \rangle \cdot \langle -\sin t, \cos t \rangle \, dt = \\ &= \int_0^{2\pi} (\sin^2 t + \cos^2 t) \, dt = \int_0^{2\pi} 1 \, dt = 2\pi \text{ units of work.} \end{aligned}$$

Notice that the circulation is positive. The reason for this is that the orientation of C "flows" with the direction of $\vec{F}$. At any point along the circle, the tangent vector and the vector from $\vec{F}$ form an angle of less than 90° , and therefore the corresponding dot product is positive.

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Table of Contents

1 6.1. Vector Fields

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3 6.3. Conservative Vector fields: The Fundamental Theorem
for Line Integrals. Independence of Path.

4 6.4. Green's Theorem

Suppose that $\vec{F}$ is a continuous vector field in some domain D .

A path C is called **closed** if its initial and final points are the same point.

A path C is **simple** if it doesn't cross itself.



Simple
not closed



Simple
closed



Not simple
closed



Not simple
not closed

A region D is **open** if it doesn't contain any of its boundary points.

A region D is **connected** if we can connect any two points in the region with a path that lies completely in D .

A region D is **simply-connected** if it is connected and it contains no holes.



Simply connected regions



Connected regions that are not simply connected



A region that is not connected

The Fundamental Theorem for Line Integrals

Theorem: Suppose that C is a smooth curve given by $\vec{r}(t)$, $a \leq t \leq b$. Also suppose that f is a function whose gradient vector, ∇f , is continuous on C . Then,

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Proof

$$\begin{aligned}\int_C \nabla f \cdot d\vec{r} &= \int_a^b \nabla f(\vec{r}(t)) \cdot \frac{d\vec{r}(t)}{dt} dt = \\ &= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt = \\ &= \int_a^b \frac{d}{dt} f(\vec{r}(t)) dt = f(\vec{r}(b)) - f(\vec{r}(a))\end{aligned}$$

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Example:

Let $\vec{F}(x, y) = \langle 2x, 4y \rangle$, if $\vec{F} = \nabla f$, where $f(x, y) = x^2 + 2y^2$. Calculate

$$\int_C \vec{F} \cdot d\vec{r}$$

where C is the line segment from $(0, 0)$ to $(2, 2)$.

Solution:

First let's notice that we didn't specify the path for getting from the first point to the second point. The reason for this is simple. The theorem above tells us that all we need are the initial and final points on the curve in order to evaluate this kind of line integral.

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(2, 2) - f(0, 0) = (4 + 8) - (0 + 0) = 12.$$

Notice that we also didn't need the gradient vector to actually do this line integral.

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The Fundamental Theorem for Line Integrals has two important consequences:

- 1 The first consequence is that if $\vec{F}$ is conservative and C is a closed curve, then the circulation of $\vec{F}$ along C is zero—that is,

$$\oint_C \vec{F} \cdot d\vec{r} = 0.$$

- 2 The second important consequence of the Fundamental Theorem for Line Integrals is that line integrals of conservative vector fields are independent of path — meaning, they depend only on the endpoints of the given curve, and do not depend on the path between the endpoints:

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

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$$\oint_C \nabla f \cdot d\vec{r} = 0$$

Theorem 2: The $\int_C \nabla f \cdot d\vec{r}$ is independent of path in D if and only if

$\int_C \nabla f \cdot d\vec{r} = 0$ for every closed path in D .

Theorem 3: Suppose $\vec{F}$ is a vector field that is continuous on an open connected region D . If $\vec{F}$ is independent of the path in D , then $\vec{F}$ is a conservative vector field on D ; that is, there exists a function f such that

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Conservation of Energy

Let's apply the ideas of this lesson to a **continuous force field** that moves an object along a path given by $\vec{r}(t)$, where t_0 is the initial point and t_1 is the terminal point of $\vec{r}$. According to Newton's Second Law of Motion, the force at a point on $\vec{r}$ is related to the acceleration by the equation

$$\vec{F}(\vec{r}(t)) = m \cdot \vec{r}''(t)$$

So the work done by the force on the object is

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_a^b m \vec{r}''(t) \cdot \vec{r}'(t) dt = \\ &= \frac{m}{2} \int_a^b \frac{d}{dt} [\vec{r}'(t) \cdot \vec{r}'(t)] dt = \\ &= \frac{m}{2} \int_a^b \frac{d}{dt} |\vec{r}'(t)|^2 dt = \frac{m}{2} \left[|\vec{r}'(t)|^2 \right]_a^b = \frac{m}{2} \left(|\vec{r}'(b)|^2 - |\vec{r}'(a)|^2 \right) \end{aligned}$$

Conservation of Energy

Let's apply the ideas of this lesson to a **continuous force field** that moves an object along a path given by $\vec{r}(t)$, where t_0 is the initial point and t_1 is the terminal point of $\vec{r}$. According to Newton's Second Law of Motion, the force at a point on $\vec{r}$ is related to the acceleration by the equation

$$\vec{F}(\vec{r}(t)) = m \cdot \vec{r}''(t)$$

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Conservation of Energy

Therefore

$$W = \frac{1}{2}m|\vec{v}(b)|^2 - |\vec{v}(a)|^2$$

where $\vec{v} = \vec{r}'(t)$ is the velocity.

The quantity $\frac{1}{2}m\|\vec{v}(t)\|^2$, that is, half the mass times the square of the speed, is called the **kinetic energy** of the object. Therefore we can rewrite the equation above as

$$W = K(B) - K(A),$$

which says that *the work done by the force field along C is equal to the change in kinetic energy at the endpoints of C .*

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Conservation of Energy

Now let's further assume that $\vec{F}$ is a conservative force field; that is, we can write $\vec{F} = \nabla f$. In physics, the **potential energy** of an object at the point (x, y, z) is defined as $P(x, y, z) = -f(x, y, z)$, so we have $\vec{F} = -\nabla P$. Then by the Fundamental Theorem for line integrals we have

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = - \int_C \nabla P \cdot d\vec{r} = \\ &= -P(\vec{r}(B)) + P(\vec{r}(A)) = P(A) - P(B). \end{aligned}$$

Comparing this equation with equation $W = K(B) - K(A)$, we see that

$$P(A) + K(A) = P(B) + K(B)$$

which says that if an object moves from one point A to another point B under the influence of a conservative force field, then the sum of its potential energy and its kinetic energy remains constant. This is called the **Law of Conservation of Energy** and it is the reason the vector field is called *conservative*.

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- 3 6.3. Conservative Vector fields: The Fundamental Theorem
for Line Integrals. Independence of Path.
- 4 6.4. Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous first-order partial derivatives on an open region that contains D , then:

$$\int_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D (Q_x - P_y) dA$$

Example 1: Evaluate the line integral

$$\oint_C y^2 dx + 3xy dy$$

if the boundary is the semiannular region D in the upper half-plane between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$

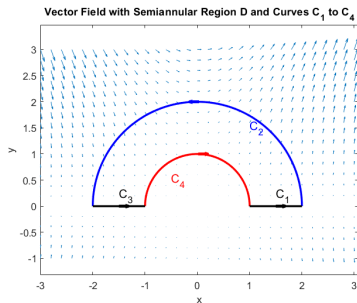
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Solution:

Before we go any further, we must decide how to evaluate, as there are two choices. If we choose to evaluate using a line integral, we will need to represent as a piecewise-smooth curve, such that:

$$\int_C Pdx + Qdy = \int_{C_1} Pdx + Qdy + \int_{C_2} Pdx + Qdy + \int_{C_3} Pdx + Qdy + \int_{C_4} Pdx + Qdy$$

It is easier to use the double integral given by Green's theorem:

$$\oint_C Pdx + Qdy = \iint_D (Q_x - P_y) dA$$

First, we will find our first partial derivatives:

$$\oint \underbrace{y^2}_{P} dx + \underbrace{3xy}_{Q} dy$$

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Next, we will notice that it would be far easier to evaluate our double integral in polar coordinates since our region is circular:

$$D = \{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$$

Now, all we have to do is integrate our iterated integral:

$$\begin{aligned} \oint_C y^2 dx + 3xy dy &= \iint_D (y) dA = \\ &= \int_0^\pi \int_1^2 r \sin \theta \cdot r dr d\theta = \frac{14}{3} \end{aligned}$$

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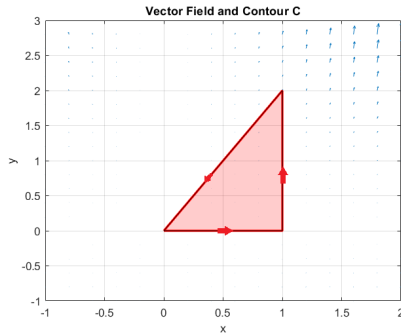
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Example 2: Use Green's Theorem to evaluate

$$\oint_C xy \, dx + x^2 y^3 \, dy$$

where C is the triangle with vertices $(0,0)$, $(1,0)$, $(1,2)$ with positive orientation.



Solution:

So, the curve does satisfies the conditions of Green's Theorem, and we can see that the following inequalities will define the region enclosed:

$$0 \leq x \leq 1, \quad 0 \leq y \leq 2x$$

We can identify P and Q from the line integral. Here they are:

$$P = xy \text{ and } Q = x^2y^3$$

So, using Green's Theorem, the line integral becomes:

$$\oint_C xy \, dx + x^2y^3 \, dy = \iint_D (2xy^3 - x) \, dA = \int_0^1 \int_0^{2x} (2xy^3 - x) \, dy \, dx$$

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Green's Theorem on General Regions

Green's theorem traditionally applies to simply connected regions, but we can extend it to regions with finitely many holes.

A boundary curve C is positively oriented if the region D is on the left as one traverses C . For a region D with holes, we divide it into two simply connected regions D_1 and D_2 with respective boundaries ∂D_1 and ∂D_2 . Assuming the outer boundary is positively oriented and the inner holes are negatively oriented, we have:

$$\begin{aligned}\oint_{\partial D} \mathbf{F} \cdot d\mathbf{r} &= \oint_{\partial D_1} \mathbf{F} \cdot d\mathbf{r} + \oint_{\partial D_2} \mathbf{F} \cdot d\mathbf{r} = \\ &= \iint_{D_1} (Q_x - P_y) dA + \iint_{D_2} (Q_x - P_y) dA = \\ &= \iint_D (Q_x - P_y) dA.\end{aligned}$$

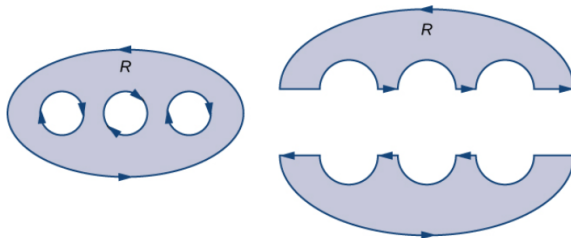
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Green's Theorem on General Regions



This shows that Green's theorem holds for regions with holes, as illustrated with an annulus where the outer boundary is oriented counterclockwise and the inner clockwise.

The resulting equation is:

$$\oint_{\partial D} \mathbf{F} \cdot d\mathbf{r} = \iint_D (Q_x - P_y) dA.$$

Green's Theorem on General Regions

Example 1: Using Green's Theorem on a Region with Holes

Calculate the integral

$$\oint_{\partial D} \left(\left(\sin x - \frac{y^3}{3} \right) dx + \left(\frac{x^3}{2} + \sin y \right) dy \right),$$

where D is the annulus given by the polar inequalities $1 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

Solution

Although D is not simply connected, we can use the extended form of Green's theorem to calculate the integral.

$$\oint_{\partial D} \left(\left(\sin x - \frac{y^3}{3} \right) dx + \left(\frac{x^3}{2} + \sin y \right) dy \right) = \iint_D (Q_x - P_y) \, dA = \iint_D (x^2 + y^2) \, dA =$$

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Green's Theorem on General Regions

Since the integration occurs over an annulus, we convert to polar coordinates:

$$x = r \cos(\theta) \quad y = r \sin(\theta)$$

The area element dA in Cartesian coordinates (i.e., $dx dy$) transforms to polar coordinates as:

$$dA = dx \quad dy = r dr d\theta$$

We can express $x^2 + y^2$ as follows:

$$x^2 + y^2 = (r \cos(\theta))^2 + (r \sin(\theta))^2 = r^2(\cos^2(\theta) + \sin^2(\theta)) = r^2$$

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Green's Theorem on General Regions

$$= \int_0^{2\pi} \int_1^2 r^3 dr d\theta = \int_0^{2\pi} \frac{15}{4} d\theta = \frac{15\pi}{2}.$$

